Ayush Kansal

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EDUCATION

University of Illinois Urbana-Champaign — *Bachelor of Science in Computer Science* Expected May 2029
- Relevant Coursework: Discrete Structures (CS 173), Linear Algebra (MATH 257), Calculus III (MATH 241).

University of California, Berkeley — *concurrent coursework* Jun 2024 – Aug 2025
- CS 61A Structure and Interpretation of Computer Programs (A–); CS 61BL Data Structures (A); Data C8 Introduction to Data Science (A).

TECHNICAL SKILLS

- **Languages:** Python, Java, C, Swift, JavaScript
- **CS Fundamentals:** data structures, algorithms, object-oriented design, discrete math, linear algebra
- **Systems & Tools:** Linux/Unix, Git, Bazel, FreeRTOS, CAN bus, Xcode
- **ML & AI:** PyTorch, NumPy, LLM integration (Claude API, Apple Foundation Models), computer vision, speech recognition

PROJECTS

- **Lotus Bud Academy Daycare Platform** (lotusbudacademy.com/software) — sole developer of a full-stack iOS, Android, and web platform in production: daily reports, photos, attendance, messaging, invoicing, and an AI assistant.
- **NoteFlash** (iOS/iPadOS; SwiftUI, SwiftData, Foundation Models) — flashcard app that turns notes, scanned pages, PDFs, slides, and Google Drive files into decks, writing cards on-device with Apple Intelligence and re-syncing changed cards from linked Drive files.
- **BTC Survivor** (Python, NumPy) — always-on service that ingests live Coinbase order-book and trade streams, turns them into features every minute, and runs 20 learning bots, with checkpoint/resume, a replay harness, a read-only dashboard, and a systemd unit; runs 24/7 on a Linux server and passed a memory soak test (paper trading only).
- **FaceUnlock** (macOS; Vision, Core ML) — fully on-device face recognition: Vision detection, 5-point alignment, Core ML embeddings with cosine matching, and blink-based liveness to reject photos, with unit-tested core logic.
- **Volt** (macOS; SwiftUI, AppKit, IOKit, CoreBluetooth) — battery menu bar app reading live current and voltage from the SMC, iPhone and iPad battery over Bluetooth LE, and AirPods levels decoded from Continuity broadcasts.

ENGINEERING EXPERIENCE

Illini Electric Motorsports — *Firmware Developer, Battery Management System* Sep 2026 – Present
- Wrote and shipped real-time C firmware on FreeRTOS (built with Bazel) for the race car's battery management system: a 1 Hz CAN heartbeat, bench-verified with a CAN bus analyzer, and a charging algorithm for the 140-cell pack that tapers current as the highest cell nears its voltage limit.

PilotCity — *Software Developer; Data Scientist* Nov 2022 – Aug 2024
- Diagnosed and resolved defects pre-deployment and optimized existing code paths; cleaned and analyzed large datasets and tightened collection methods to reduce analysis error.

Seeed Studio — *Intern* Jun – Aug 2023
- Developed AI software for edge computing on microprocessors; built an AI-powered subject-tracking camera that maintained tracking beyond frame boundaries.

Alan AI — *Intern* Jul – Sep 2022
- Built a voice-activated GitHub FAQ on Alan AI's speech recognition platform, targeting a ~30% reduction in support call wait times.

RESEARCH EXPERIENCE

ASDRP — *Lead Researcher, Author, and Developer* Jun 2024 – Jan 2026
- Built an iOS app and an AI Google Chat bot that manage profiles, lab workflows, and reminders for 30+ lab groups and 1,100+ student researchers.
- Co-first author of a paper under review at the *Journal of Emerging Investigators*; built the Python pipeline that processed 150 hours of audio across four speech recognition systems and found speed thresholds that cut transcription cost ~20%.

Osteoporosis Gait Sensor — *Lead Developer and Designer* Jun 2024 – Present
- Designed and prototyped a wearable foot sensor that uses machine learning to detect early osteoporosis indicators from gait data.
- Built the instrument around a Raspberry Pi and a 16-sensor array laid out in the shape of a foot, with a real-time collection, preprocessing, and model-based analysis pipeline.

LEADERSHIP

Byte By Byte Coding — *CEO and Co-Founder* Dec 2024 – Present
- Founded a 501(c)(3) teaching computer science to 50+ underserved students; built the curriculum and manage volunteer instructors and student progress tracking.

Fremont Debate Academy — *Lead Instructor and Director of Operations* Dec 2022 – Jun 2025
- Directed operations and coached 100+ students within a program serving 7,500; trained incoming instructors to scale the program.

Boy Scouts of America — *Eagle Scout; Senior Patrol Leader; NYLT Staff* Aug 2018 – Jun 2026
- Attained Eagle Scout; led troop activities and service projects, and taught leadership courses at National Youth Leadership Training. Earned the Presidential Volunteer Service Award (Gold) twice for 100+ service hours each.